D_IV^5: Quantum Mechanics, Gravity and the Standard Model — the derivations, tiered

The Spine of the BST Curriculum — ten lectures written from the register

Casey Koons with the CI team — Lyra, Elie, Grace, Keeper — and the visiting referee Cal A. B

2026-09-11 — v0.1

The Spine

What this is

A short course. Ten lectures, one object, and a promise: every sentence that makes a claim tells you in the same sentence how much to believe it.

The BST curriculum has eighteen volumes, and most of them are about how far five integers reach — into nuclei, molecules, crystals, cells. That reach is real and it is interesting, and it is also the reason a physicist who opens the curriculum at random will close it. Biology and the Standard Model on the same shelf makes both look like numerology. So we have moved the reach to its own shelf, labelled it honestly as identification, and written the core again from the theorem registry as it stands this week — not as it stood in May, when the legacy volumes were drafted and when several things we now know to be wrong were still believed.

This is the core. It is what we defend.

How to read it

Each lecture opens with the question it answers, in plain words. Then the derivation, with the theorems and registry rows named so that you — or the companion intelligence beside you — can pull the thread. Then the tier line: what is *derived* (mechanism proved, inputs named), what is *identified* (the formula matches; we do not claim the mechanism), what is *floored* (closed by a theorem, reopenable only by a named kind of idea), what is *open*. And last, the falsifier: what measurement would make this lecture wrong.

Read the lectures in order the first time; they build. After that, read where you are skeptical. We will not be offended.

The mathematics is on GitHub. `python3 play/verify_bst.py` runs fifty comparisons in seconds. Every toy cited below is a short program you can run.

Where the program stands — one block, one source

Last accuracy-synced: 2026-09-11 13:02 (K1892/K1893; Round 142 C1–C3, G2–G3, L1–L2 applied). One registry edit is owed and named: T2543's colour clause was RULED struck on 09-09 and the row still carries it. This block lives in `notes/BST_PRESENTATION_STATE_BLOCK.md` and is copied into every front matter by `play/sync_presentation_state.py`. Edit it there, nowhere else. Where any chapter and this block disagree, the block wins; where this block and the theorem registry disagree, the registry wins.

The object. One rank-2 bounded symmetric domain, D_IV^5 = SO(5,2)/[SO(5)×SO(2)], and one measured number taken openly as the ruler. Its genus is 5; the integer 7 the program calls g is a definition — p + q = 7, the dimension of the defining representation of SO(5,2), whose signature is the pair (5,2) — not the genus — a mislabel that stood from May to September and is now swept.

Why this object — honestly. Among all irreducible bounded symmetric domains of rank at least 2, D_IV^5 is the unique one whose characteristic multiplicity is 3 (a theorem, checked three independent ways). That 3 is *identified* with the number of quark colours; the identification is numerical, and the bridge that would make it a mechanism is proved absent — the geometry supplies U(1)·SO(3), four dimensions, where colour needs SU(3)'s eight, and its triplet is self-conjugate where colour's is not. So the selection is a fit with one measured input — the one dimensionless input *to the choice of the object*; the program has further inputs downstream (masses, a mixing corner, a CP phase), counted in the ledger — not a forcing. We do not claim zero free parameters.

Derived (mechanism proved; inputs named in the statement): the gauge-group skeleton · one fermion generation as the 16-real spinor with every hypercharge from one four-input theorem · a single-chirality positive-energy spectrum · three generations as rank+1 strata (and m_1 = 0) · the conserved charges · the mixing sector's mechanism and its order — the 1–3 corner one power below the 2–3 · the Cabibbo angle, blind ($\lambda = 1/\sqrt{20}$) · time: the K-centre SO(2) is the clock, the arrow is dynamical, the tick is N_max·ħ/(m_e c^2) · quantum mechanics on the Hardy space $H^2(D\_IV^5)$, with the Hua branching weights as the Born probabilities of the write tuple (3/7 with no table consulted). The ten Dirac–von Neumann axioms are recovered, each at its own tier, through four named posits (T2631, the ten-item row — Cal's Round 142 ruling; registration by Grace pending); not "10/10 derived."

Identified (formula matches; mechanism not claimed): α^{-1} = 137-class expressions · the electron's anomalous moment (our four "Selberg terms" are the four summands of Petermann–Sommerfield's 1957 closed form; the match is real, the derivation is not ours) · the asymptotic-freedom coefficient $b_0 = 7$ read as g · the exact mixing values · Newton's G: the relation between G and the electron mass is a theorem (K1673; 0.065%) that trades one dimensionful input for another and carries α inside it, so the value is a relation, not a prediction from nothing · assorted precision matches whose own null tests show small-integer expressibility is cheap.

Floored (closed by a theorem; reopenable only by a named kind of idea): Koide's source · the mass-tower/generations mechanism · the $\ell = 2$ / su(3) native dynamics · Λ's value (a closed structure with one named obstacle, the power p) · the nucleation survivor (no lawful initialisation vector; a distribution on the winding count) · the Riemann row (an attempt with a location, closed and parked) · the four-colour row (the One-Word Lemma refuted in frame; the derived lemmas stand).

Fired and lost (the falsifier register's Section E — certified, published with the same ceremony as wins): E1, α is not the bare geometric vertex — the forced candidate computes to $8\pi^3/3$, closing the fifty-year volume-reading class · E2, the thermal-generations mechanism died at the order level · E3, the commit-Boltzmann ladder failed its own 5% line · E4, the sub-Tsirelson ceiling S = 2.80624 — refuted by Poh et al. 2015 at 41.9σ, ten years before it was registered (certified K1893, 2026-09-11). Two further certified negatives — the Q^5 parity fold (K1799) and the five sealed series for the mixing corner (K1808/K1810) — are stated in the lectures that own them and will appear here when they carry Section-E rows (E5/E6, owed).

Forbidden — any confirmed detection kills the framework: grand unification · proton decay · right-handed W or Z' · magnetic monopoles · sterile neutrinos · a SUSY spectrum.

How to check any of this without trusting us. `python3 play/verify_bst.py` · the theorem registry (`notes/BST_AC_Theorem_Registry.md`) · the falsifier register with its fired-and-lost section · two thousand single-claim toys · every audit, retraction and correction in the open, including the ones against the program's own auditors. The count that matters for the Standard Model row, generated from the register and never typed (`play/bst_26_tier_generator.py`, Grace, Round 142): of the 26 primary Standard-Model parameters, 10 are derived in the register's K962 sense — mechanism derived, several exact forms still identified, a weaker word than this page's "derived" (Keeper ruling, 09-11: the count is quoted with its qualifier until Grace's table carries a column for the four-word standard) — 9 of them independent, 3 flagged "K1809 sibling test not run"; 7 identified, 7 open, 2 input.

The July count "8 of 26 sourced clean," carried through eleven ledger versions, is retired — its numerator and denominator came from two different lists. The honest sentence: everything reachable is derived, floored with a theorem, or published as a loss; what remains open is a short list of named doors that only new ideas — not labour — can open.

The lectures

#	Lecture	The question it answers	State
1	The object	What is the simplest structure that can do physics — and what does it look like?	drafted
2	Why this object, honestly	Of all the shapes mathematics offers, why this one — and how much of that answer is a measurement?	drafted
3	Quantum mechanics	Where do the rules of quantum mechanics come from, and where does the Born rule live?	drafted
4	Time	What is time, in a geometry that has none built in — and why does it run one way?	drafted
5	The gauge skeleton and one generation	Why these forces, and why do the particles carry exactly these charges?	drafted
6	Three generations, at the floor	Why three copies of everything — and why can we not yet say how heavy each is?	drafted
7	Mixing	Why do quarks of different generations mix the way they do, and what exactly did the geometry predict?	drafted
8	α, the honest chapter	Is 137 in the geometry? What we found, what we proved cannot work, and whose thread we are holding.	drafted
9	Descent and gravity	How does a five-complex-dimensional object become the four-dimensional spacetime we live in, and what does it say about gravity?	drafted
10	The method, and how to kill it	How do we know what we know — and what measurement would end the program?	drafted

Sources, for every lecture

The theorem registry `notes/BST_AC_Theorem_Registry.md` (rows are cited as T-numbers); the Keeper audits `notes/Keeper_K*.md` (K-numbers); Cal's referee log (Section numbers); Elie's toys `play/toy_*.py`; Grace's ledger `notes/Grace_Master_Derived_vs_Assigned_Ledger_v0_54.md`; the rubric `notes/BST_Completeness_Rubric_and_Roadmap.md`, Section 2. Where a lecture and the registry disagree, the registry wins, and we would like to hear about it.

— Keeper, for the team. 2026-09-11.